

# QANDEEL

## قندیل

إدارة أسراب قناديل البحر وإعادة إدخالها إلى البحر حية

Herd, don't harvest: autonomous, live-release management of Blue Blubber jellyfish swarms after ENEC's early warning

### Team Qandeel

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# We keep swarms out of the intake and return them to the sea alive

## Problem

Blue Blubber swarms (30–45 cm, 1–3 kg) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

## Solution

After ENEC’s alert, a drift planner decides when to act; a bubble curtain at the breakwater gap holds the swarm out; two uncrewed boats tow a boom with a closed-bottom bag and release the jellyfish alive. An AI gap camera is the backup.

## Evidence

Simulation on real 2025 Gulf currents and wind, stress-tested against physics the planner does not know; online dashboard; 26 tests; open code.

## Next

A 1:20 tank test (AED 300 kit) to measure what the model assumes, then a sea pilot with ENEC.

**3% vs 23%**

of arriving jellyfish reach the gap before the curtain runs: Qandeel vs a daylight camera alone (stress test, 3 × 100 alerts, real 2025 data)

**1–2 days**

of warning for the boats and the curtain, for about 12 extra curtain hours (~USD 500 of energy) per alert

**84–100%**

of jellyfish kept by the boom with our retention bag (model), vs 0–55% for an open oil-spill boom

**29–46%**

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind (variable-speed compressor)

# A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work, less harm and low energy, in line with the UAE Net Zero 2050 and biodiversity goals.

### The target: Blue Blubber

- One of the two dominant outbreak species on the UAE coast; reproduces year-round (Aravind 2024)
- 30–45 cm bell, ~1–3 kg each (Yang et al. 2018 mass formula)
- 87–100% sit in the top 1 m by day (Pitt & Kingsford)
- Large adults swim >10 cm/s: not just passive drifters

### What happens at intakes

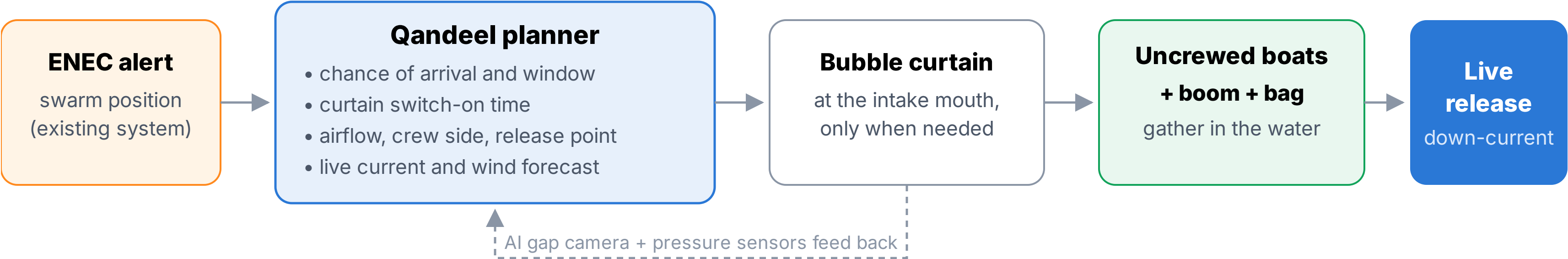
- Swarms follow the intake flow through the breakwater opening and pile on screens
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013)
- In the region, ~300 t of jellyfish clogged intakes in Oman

### Options today, compared

| OPTION                | JELLYFISH             | DRAWBACK                            |
|-----------------------|-----------------------|-------------------------------------|
| Rake screens (now)    | die                   | reactive, heavy labour, outage risk |
| Fixed nets offshore   | die, bycatch          | clog in big blooms                  |
| Cutting robot (JEROS) | shredded              | ~0.9 t/h, kills                     |
| Harvest for food      | used                  | no UAE market yet                   |
| <b>Qandeel</b>        | <b>released alive</b> | unproven at sea: tank, then pilot   |

No management method has been tested on this species.

# Guard the mouth, move the water, keep them alive



## 1 • The mouth is a chokepoint

At Barakah, seawater reaches the pumps through one ~3 km channel. We measured its mouth on satellite images: 967 m.

## 2 • Move the water, not the animal

A rising-bubble current across the mouth holds jellyfish out at sea; nothing is cut, pumped or netted.

## 3 • Proven tools, made autonomous

Oil-spill booms on uncrewed boats, plus a bag that stops leaks. Release west, away from the discharge.

# One swarm, from alert to live release

- 1

Alert.

ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
- 2

Plan.

400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 44% chance of arrival, window 16–73 h.
- 3

Prepare.

Curtain on at +13 h; uncrewed boats staged on the approach side; release point chosen.
- 4

Deflect.

Curtain across the gap holds the swarm outside; airflow follows the current.
- 5

Gather.

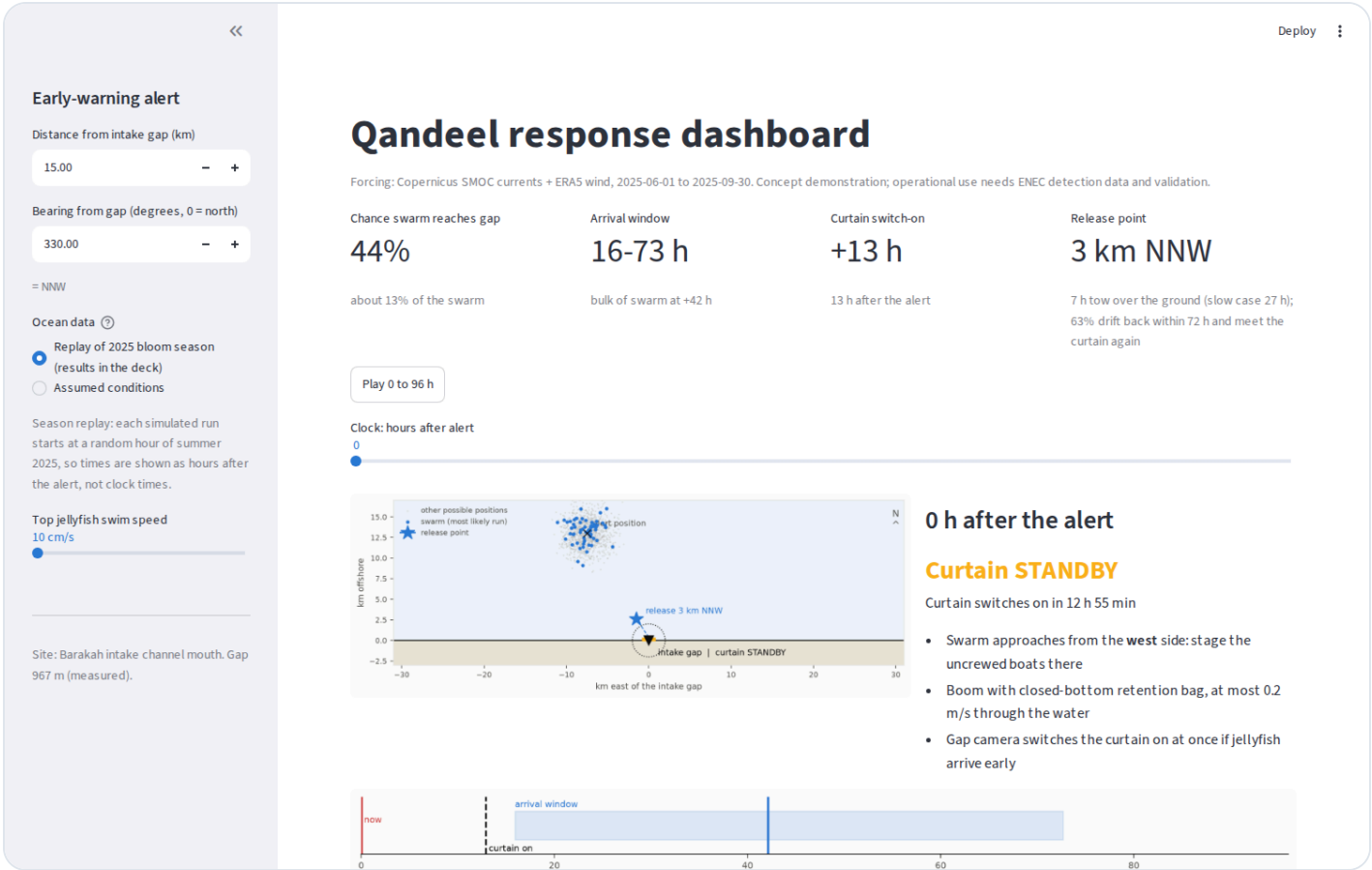
Two uncrewed surface vessels, supervised from shore, close a U-boom with a closed-bottom bag.
- 6

Release.

Tow ~3 km NNW ( $\leq 0.2$  m/s through the water: ~7 h over the ground, up to a day against strong inflow) and open the bag; ~60% drift back and are held again.
- 7

Learn.

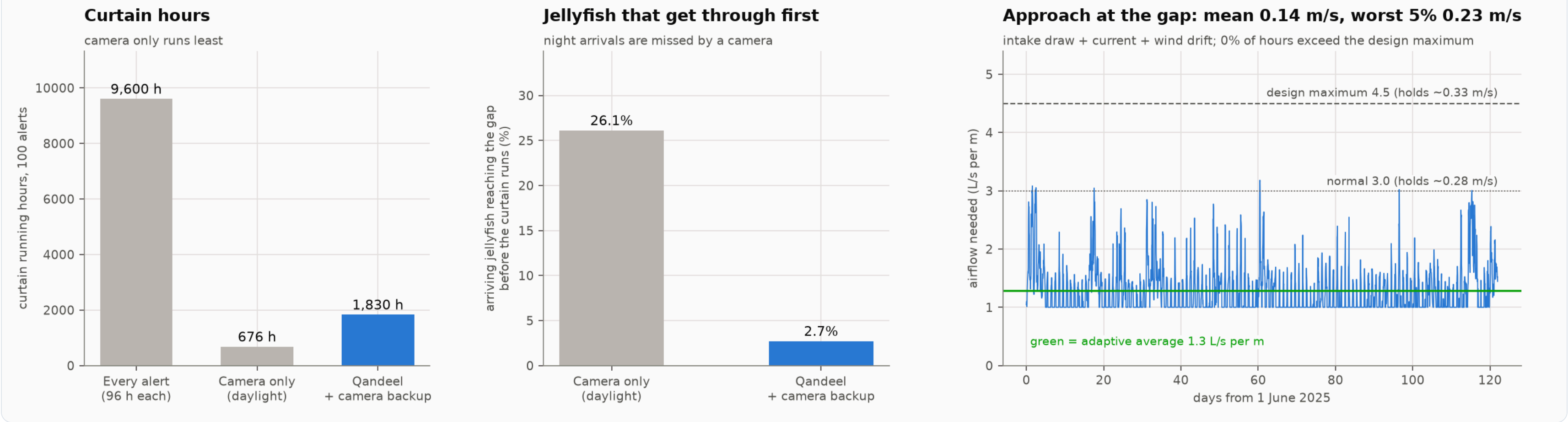
An AI camera counts jellyfish at the gap and starts the curtain if they arrive early. Its records are the training data for a learning model that corrects the drift forecast.



Qandeel operator dashboard (working prototype, Streamlit): chance of arrival, window, switch-on clock time, swarm map, curtain status, release point. Data: Copernicus SMOC currents + ERA5 wind, summer 2025.

# Better protection than a camera alone, at a known cost

Model-error stress test, 100 alerts in the real 2025 season: Qandeel lets 2.7% through vs 26% for a camera alone



100 alerts at random times and places in the real 2025 season; the truth uses physics the planner does not know. Base case 1% wind drift. "Arrived" = within 2 km of the gap (15 km NNW alert: 44%; within 500 m: 31%).

### 3% vs 23% get through

Truth: currents  $\times 0.7\text{--}1.3$ , wind drift  $\times 0.5\text{--}1.7$ , an unmodelled 0–3 cm/s current. Three random seeds: 2.7–3.7% vs 20–26%. A daylight camera misses night arrivals.

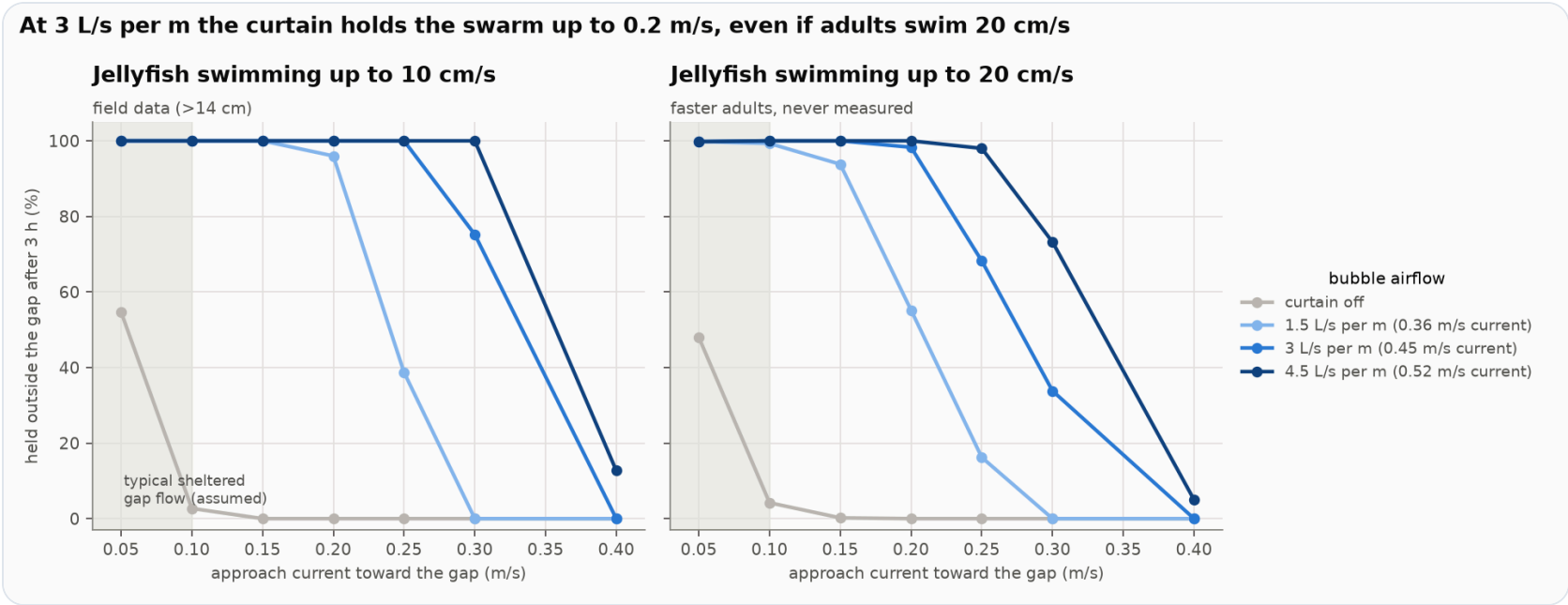
### The cost

~1,800 curtain hours vs ~700 for camera-only (~12 h, ~USD 500 of energy, per alert); ~80% fewer than running on every alert. In return: 1–2 days of warning.

### What this is not

Still simulation against simulation, not real jellyfish. Next: compare drift with real Gulf drifter tracks and ENEC swarm records.

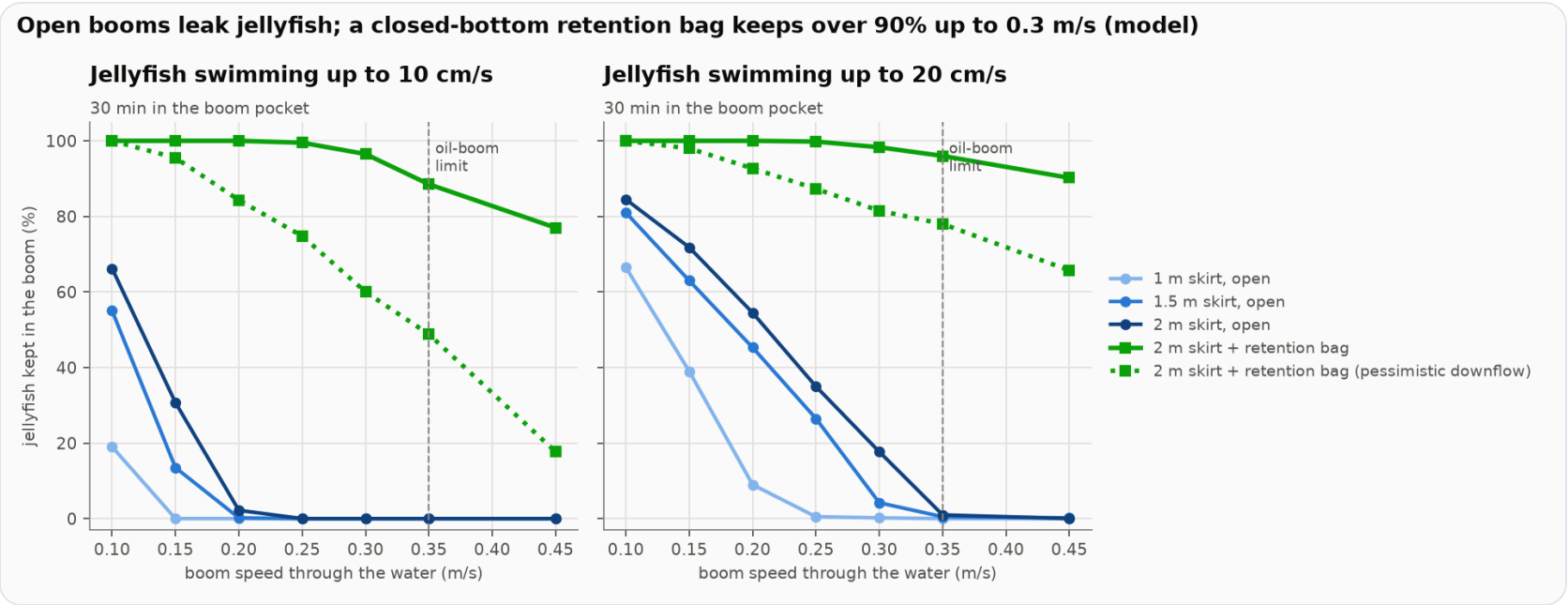
# Our finding: oil-spill booms leak jellyfish, and a bag fixes it



400 individually simulated jellyfish per point; surface current from Bulson's pneumatic-barrier relation.

### Curtain holds, if adults swim slowly

Water approaches the gap at 0.14 m/s on average, 0.23 m/s in the worst 5% of hours (0.20 / 0.36 if wind drift is 3%). 3 L/s per m holds ~0.27 m/s; 4.5 holds ~0.33. Over a full 82 h event: 100% held at 0.2 m/s, but 78% if adults swim 20 cm/s.



Jellyfish depth from field data; apex downflow 25% (pessimistic 50%) of tow speed.

### Why the bag matters

Oil floats back up; neutrally buoyant jellyfish are dragged under the skirt far below the 0.35 m/s oil-boom limit (0–55% kept at 0.2 m/s). A closed-bottom fabric bag keeps 84–100% in the model.

# Gentle on marine life, built from parts that exist today

## Marine-life protection

| RISK                                 | HOW WE REDUCE IT                                                                                                                                                                |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bubbles drawn into the intake        | Curtain on the seaward side; bubbles surface in under a minute, in the outer basin; bubble detector on the intake line; curtain moved further out if the pilot shows carry-over |
| Nuclear approval, security           | FANR approval from day one; kit stays outside safety-related intake structures; boats operated by plant staff inside the security zone rules                                    |
| Jellyfish harmed                     | No cutting, pumping or mesh; coarse bubbles push them away at the surface (air under the bell checked in the tank test)                                                         |
| Turtles, dolphins, dugongs, seagrass | Escape gap, slow tow, camera stop on sighting; pipe routed away from seagrass and coral; release points kept out of protected areas                                             |
| Released jellyfish strand            | Left alone, 55% of the same swarm strands within 4 days; release 3 km out strands 29%                                                                                           |
| Sediment, bubbly water               | Diffusers 0.5–1 m above the bed; turbidity watched at the intake; no boats or divers in the bubble zone while it runs                                                           |

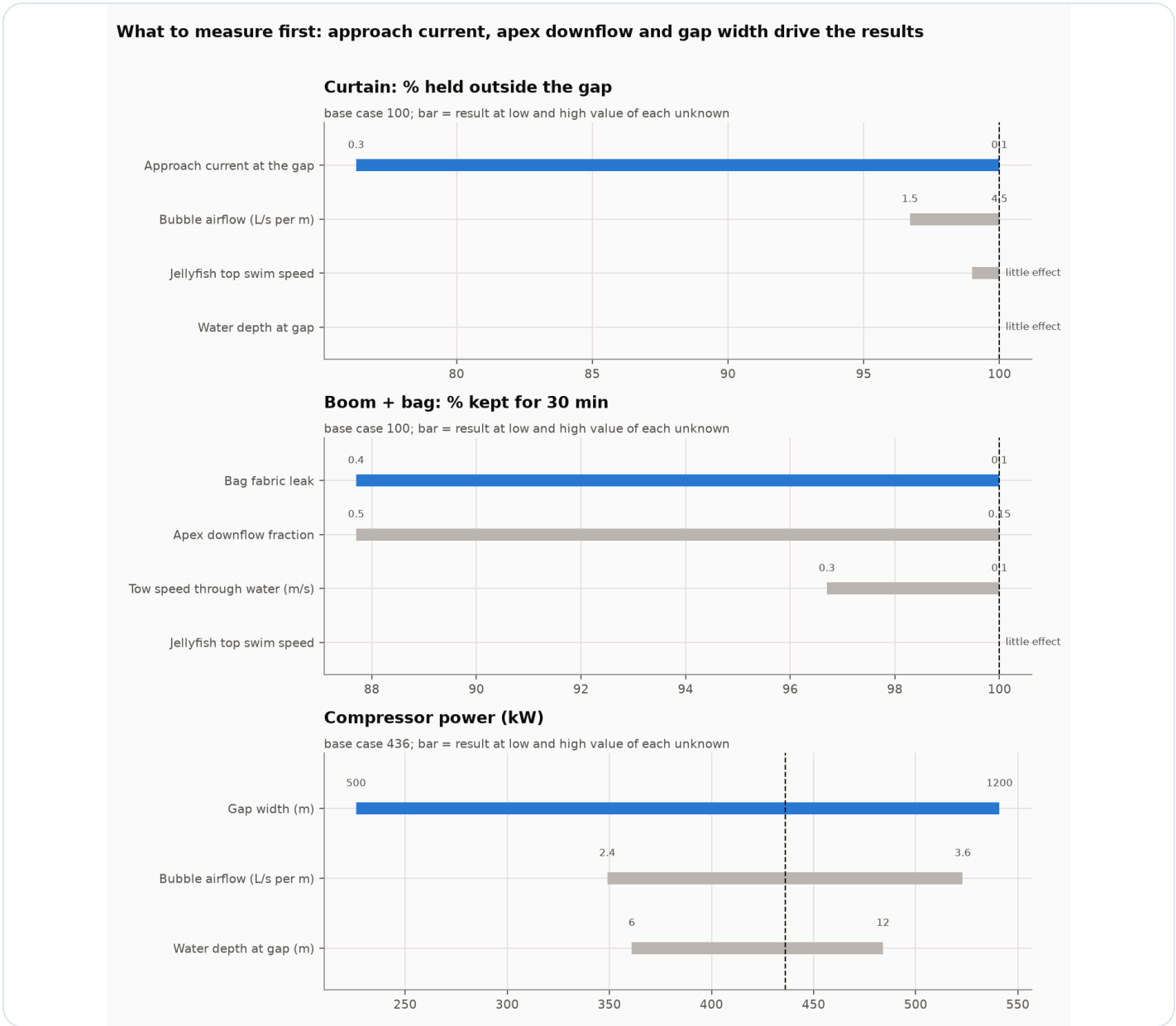
## Feasibility

| ITEM                 | ESTIMATE (967 M MOUTH)                                                                                                                                                                                           |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compressed air       | 2.9 m³/s, ~440 kW for the 967 m mouth, oil-free (ISO 8573-1 Class 0)                                                                                                                                             |
| Capital (indicative) | USD 1.2–3.5 M: 967 m curtain, variable-speed compressor + standby, boom, bags, two uncrewed boats, sensors                                                                                                       |
| Running cost         | USD 0.18–0.45 M a year: operators on call, boat upkeep, diver cleaning, servicing, ~USD 3,100 energy per event (~31 MWh: 12.6 t CO₂ on a gas grid, 0.4 t on the plant’s own power)                               |
| Value                | Two to seven avoided 12 h unit outages (~USD 0.5 M each, EPRI figure) repay the capital; under one a year covers running costs                                                                                   |
| Capacity             | Gathering 4–43 t/h, but the loaded tow against the inflow sets the pace: ~0.4 t/h net per boom unit, ~1.4 t/h with three uncrewed tugs swapping bags. The curtain does the bulk work; boats clear what builds up |
| Gulf hardening       | Membrane diffusers, foul-release coating, modular sections, lifted out off-season                                                                                                                                |

# What we assume, and what we measure first

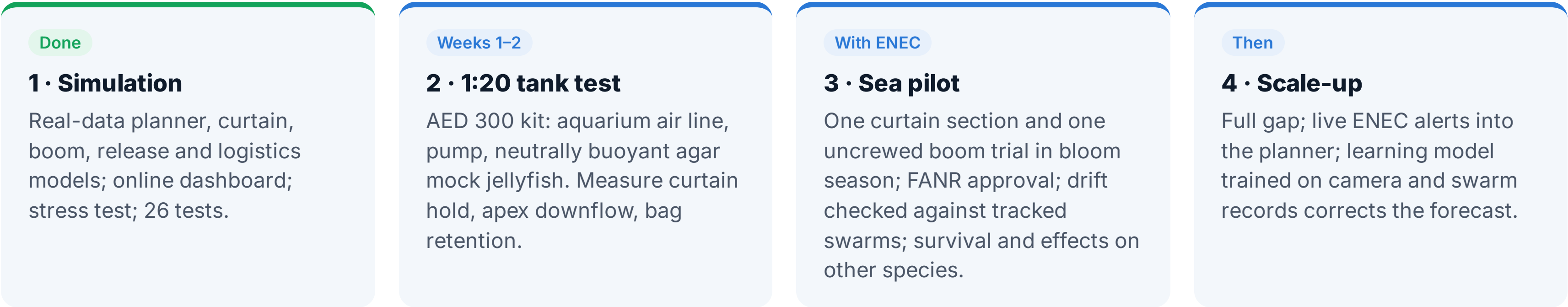
| ASSUMPTION                                              | BASIS                                     | IF WRONG                                                                     |
|---------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------|
| Intake mouth 967 m wide (measured), ~9.5 m deep         | Satellite images; public channel depth    | Depth to survey; a shorter curtain inside the channel would cut air and cost |
| Alert gives position, hours of notice                   | Brief: "time to prepare"                  | Camera trigger switches the curtain on                                       |
| Swarm in top 1 m; adults swim $\leq 10\text{--}20$ cm/s | Field and lab studies (Australia)         | Deeper skirt; tested at 20 cm/s                                              |
| Swarm density 0.1–1 per m <sup>3</sup>                  | No published figure                       | Capacity scales linearly                                                     |
| Extra wind drift 1% of wind (0–3%)                      | Ocean model already has wind-driven drift | 28 km N alert: 3–29% arrival; curtain sized for 3%                           |
| Apex downflow 25–50% of tow speed; bag passes 20%       | Model hypothesis                          | Measured in the tank test                                                    |

**Limitations:** tested only against simulations; currents from one model point, rounded to ~0.03 m/s by the free API (direct Copernicus download ready); no depth: the curtain draws water in near the bed, so deep or night-time jellyfish could pass under; species at Barakah and adult weight to confirm. **Why not harvest?** A full bag can go to a licensed processor; release is the default while no UAE market exists.



Sensitivity: approach current, bag leakage / apex downflow and gap width drive the results, so we measure them first.

# From simulation to sea, one measured step at a time



## Pass marks

- Tank:  $\geq 95\%$  held at the scaled 0.2 m/s current
- Tank: bag keeps  $\geq 90\%$  at the scaled 0.2 m/s tow
- Pilot: no harm to non-target species; fewer screen-clearing hours

## Open work and AI use

Live dashboard: [enechackathon.streamlit.app](https://enechackathon.streamlit.app). Code, data and tests: [github.com/Shaaah-7/ENEC](https://github.com/Shaaah-7/ENEC). AI assistants helped search and summarise literature, check claims, draft text and write simulation code, and an open text-to-speech model voices the video; the team reviewed every decision, assumption and figure, and can explain each in our own words.

